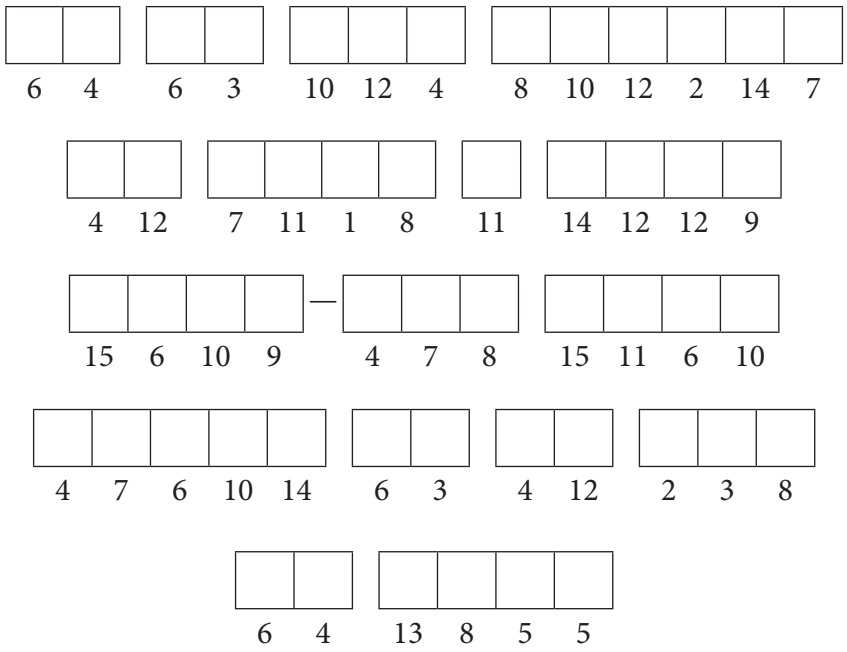
Name: _____

WORKSHEET

Instantaneous rates of change



René Descartes

Rule

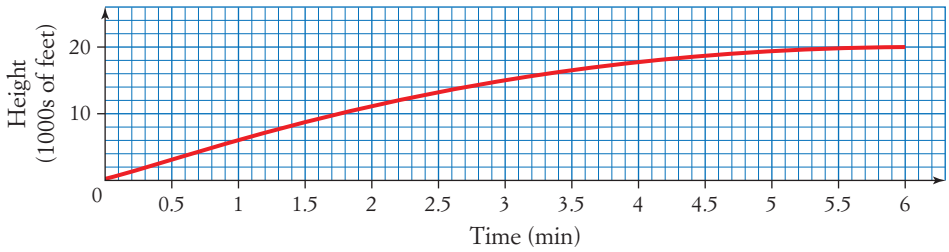
To find the instantaneous rate of change at a point x on a curve, draw a tangent at that point. The gradient of the tangent is equal to the instantaneous rate of change at that point. To find the gradient of the tangent, choose two points on the line, (x_1, y_1) and (x_2, y_2).

$$\text{Gradient of the tangent} = \frac{y_2 - y_1}{x_2 - x_1}$$

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Questions

A plane takes off and ascends to 20 000 feet as shown in the graph below.



Find the instantaneous rate of change at:

1 1 min

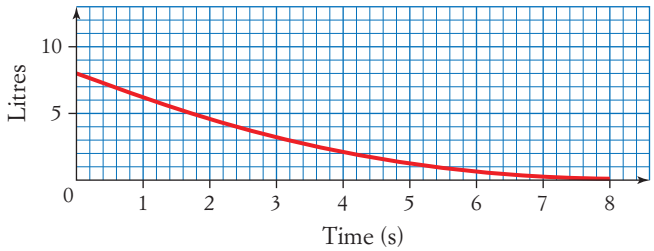
2 2 min

3 3 min

4 4 min

5 5 min

An 8 L watering can is emptied as shown in the graph below.



Find the instantaneous rate of change at:

6 1 s

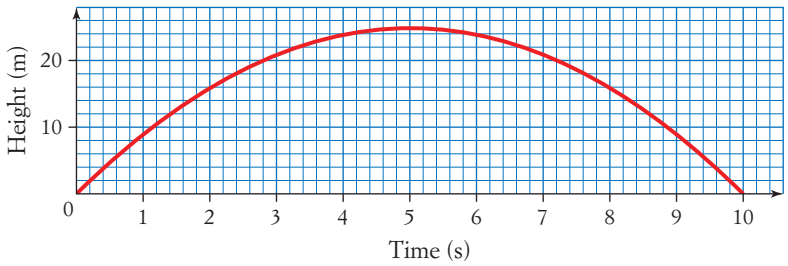
7 2 s

8 3 s

9 5 s

10 7 s

A ball is thrown into the air and its height at various times is shown in the graph below.



Find the instantaneous rate of change at:

11 1 s

12 4 s

13 5 s

14 7 s

15 8 s

Solutions

A 8 m/s

D $-\frac{3}{4}$L/s

E $-1\frac{1}{4}$L/s

G -4 m/s

H $-1\frac{1}{2}$L/s

I $-1\frac{3}{4}$L/s

L 500 feet/min

M -6 m/s

N $-1\frac{1}{4}$L/s

O 2 m/s

S 4000 feet/min

T 1000 feet/min

U 5000 feet/min

V 6000 feet/min

W 0 m/s

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